METRIC

A-A-50433A

August 24, 2021

SUPERSEDING

A-A-50433

August 15, 1989

COMMERCIAL ITEM DESCRIPTION

GREASE, SEAWATER WASH RESISTANT

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) covers seawater wash resistant grease. This grease has replaced MIL-PRF-24139 for use on board submarines.

2. SALIENT CHARACTERISTICS.

2.1 Requirement. Grease supplied for this CID shall be in accordance with the requirements of S9510-AB-ATM-010/(U) and T6350-AA-HBK-010, and shall meet the testing requirements and methods in table I (ASTM test procedures are listed where applicable).

TABLE I. Chemical and physical requirements and test methods.

Property	Requirement	Test method
Worked penetration at 25 °C (77 °F), 60 strokes	240 to 295	ASTM D217
Prolonged worked penetration after 100,000 strokes, maximum	295	ASTM D217
Maximum apparent viscosity	30 Pascal-seconds at 200 seconds^{-1} shear rate and 0 °C (32 °F)	ASTM D1092
Dropping point °C minimum	260 (500 °F)	ASTM D2265
Gelling agent	Aluminum complex	N/A
Percent alkali, maximum	0.01	ASTM D128
Percent water, maximum	0.01	ASTM E203
Free fatty acid percent, maximum	0.01	ASTM D128
Oxygen stability, 345 kilopascals (50 pounds per square inch) at 100 °C (212 °F) for 100 hours	Maximum drop 28 kilopascals (4 pounds per square inch)	ASTM D942
Water resistance, maximum sprayoff	≤65.0 percent	ASTM D4049
Corrosion on copper, maximum	1B	ASTM D4048
Operating range	-9 to 177 °C (15 to 350 °F)	N/A

Beneficial comments, recommendations, additions, deletions, clarifications, and any other data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

FSC 9150

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TABLE I. Chemical and physical requirements and test methods – Continued.

Property	Requirements	Test method
Elastometer compatibility with Nitrile Butadiene Rubber – Low ACN (NBR-L) and fluorocarbon (FKM)		ASTM D4289
Maximum volumetric change, percent	≥0 and ≤+10	
Maximum hardness change	≤+5 durometer points	
Steel on steel wear	≤1.25-millimeter (0.050 inch) wear scar after 90 minutes at 60 cycles per minute at 14 kilograms (30 pounds) load with 90 degree angle of oscillation, H-30 Test Block and S-10 Test Ring	ASTM D3704
Pumpability performance		NADC report serial 4123/6061 ser 007131, dated 17 Aug 1989
Pump Test Assembly @ 23.3 °C (74 °F), minimum	2.55 grams/minute (9.0 ounces/minute)	
Pump and Backpressure and Stay Time (BST) Test Assembly @ 23.3 °C (74 °F), minimum	1.7 grams/minute (0.6 ounce/minute)	
Pump and BST Test Assembly @ 1.1 °C (34 °F), minimum	0.57 gram/minute (0.2 ounce/minute)	
Two-week (336-hour) Stay Time @ 1.1 °C (34 °F), minimum	0.57 gram/minute (0.2 ounce/minute)	
2000-hour Stay Time @ 1.1 °C (34 °F), minimum	0.57 gram/minute (0.2 ounce/minute)	

2.2 Material. The grease shall consist of a liquid lubricant and gelling agent that when formulated results in a compound meeting this CID. The grease shall not contain molybdenum disulfide additive.

2.3 Prohibited materials. The grease shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

2.4 Toxicity. When evaluated in accordance with 3.3, the grease shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 3.3 and 5.2).

2.5 Off-gassing. The grease shall be evaluated for off-gassing in accordance with the requirements of 3.4. Based on the circumstances of use and the chemical nature of the grease, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the grease shall be assigned to either “Permitted” or “Limited” category (see 3.4 and 5.3).

2.6 Shelf life. The product provided shall meet the minimum shelf life of 3 years when stored under normal conditions in a sealed container. Storage of containers shall be under cover to avoid exposure to weather and temperature variations.

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3. PRODUCT CONFORMANCE PROVISIONS.

3.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

3.2 Prohibited materials. Prohibited materials shall be verified as specified (see 2.3 and 5.4).

3.3 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 2.4 as specified. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the grease using data provided by the manufacturer/distributor to the NMCPHC (see 2.4 and 5.2).

3.4 Off-gassing. The grease shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 2.5 and 5.3). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority. The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 2.5).

4. PACKAGING. Preservation, packaging, and marking shall be as specified in the contract or order. The containers shall be 15.9-kilogram (35-pound) (18.92-liter [5-gallon]) cylindrical (non-tapered) cans suitable for use with pneumatic grease guns with follower plates (NSN 4930-01-223-3730).

5. NOTES.

5.1 Source of documents.

5.1.1 ASTM Standards. ASTM standards are available online at www.astm.org.

5.1.2 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil/.

5.1.3 Naval Sea Systems Command (NAVSEA) Publications. NAVSEA publications are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626. Copies of the chapter titled "Material Control Program" are available by email request to commandstandards@navy.mil.

5.1.4 Report. Copy of grease pumpability performance in NADC report serial 4123/6061 ser 007131 may be obtained by email request to commandstandards@navy.mil.

5.2 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

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5.3 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are accomplished through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, NAVSEA S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to the Naval Sea Systems Command, SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to commandstandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which is normally accomplished through off-gas testing. The off-gas test is required to be conducted in a Government approved laboratory designated by the preparing activity. Information pertaining to this test requirement may be obtained from the Naval Sea Systems Command, SEA 05S, 1333 Isaac Hull Ave., SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to commandstandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

5.4 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Requirement for prohibited materials verification (see 2.3 and 3.2).
d. Packaging requirements.
e. Quantity required.

5.5 National stock numbers (NSNs). NSNs 9150-01-306-9167 (15.9 kilogram [35 pounds] packaging) and 9150-01-352-9706 (15.9 kilogram [35 pounds] packaging) cover the grease. NSN 4930-01-223-3730 covers the pneumatic grease gun that will be used on the grease cans.

5.6 Subject term (key word) listing.

Pumpability performance
Submarine

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – CD
Navy – SH

Preparing Activity:
Navy – SH
(Project 9150-2021-002)

Review Activities:
Air Force – 68
DLA – GS, PS

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